Table 1. FF++ intra-domain evaluation — Ding layout

Our rows: seed 17 TEST, stored validation-real FPR10% thresholds; no test recalibration. Published rows retain Ding's reported numbers. Shared format does not establish identical training/evaluation protocols.

A/B/C: earlier controlled pilot. R0/R1/R2/RN/RD: completed upgrade. C is invariance-only; RN/RD use six sources. Current 50+50 results are VAL only and excluded.

Attributes	Methods	Backbone	F_FPR ↓	F_DP ↓	es-AUC ↑	AUC ↑
Gender	Ori (published)	Xception	4.10	5.72	91.93	92.69
Gender	DAG-FDD WACV'24 [20]	Xception	1.82	4.65	94.87	96.72
Gender	DAW-FDD WACV'24 [20]	Xception	0.78	9.52	95.76	97.46
Gender	PG-FDD CVPR'24 [24]	Xception	0.62	4.74	96.32	97.66
Gender	Fairadapter ICASSP'25 [10]	ViT-L/14	4.16	12.21	67.85	71.50
Gender	RSEF-FDD arxiv'25 [18]	Xception	0.57	8.55	94.91	97.09
Gender	Ding et al. (published)	Xception	0.53	3.61	96.45	97.71
Gender	CES-FL A: ImageNet frozen [s17]	Xception	2.89	6.88	65.06	67.48
Gender	CES-FL B: detection frozen + BCE [s17]	Xception	2.02	1.74	96.94	98.35
Gender	CES-FL C: detection frozen + invariance [s17]	Xception	1.99	1.82	96.73	98.22
Gender	CES-FL upgrade R0: frozen BCE [s17]	Xception	2.02	1.74	96.94	98.35
Gender	CES-FL upgrade R1: partial BCE [s17]	Xception	0.17	2.78	96.99	98.40
Gender	CES-FL upgrade R2: partial DRO [s17]	Xception	0.88	3.85	97.02	98.40
Gender	CES-FL upgrade RN: no-change diagnostic [s17]	Xception	0.26	4.04	96.67	97.85
Gender	CES-FL upgrade RD: age diagnostic [s17]	Xception	0.75	3.82	96.73	98.20
Race	Ori (published)	Xception	19.76	4.74	82.85	92.69
Race	DAG-FDD WACV'24 [20]	Xception	5.48	9.20	93.43	96.72
Race	DAW-FDD WACV'24 [20]	Xception	5.43	14.69	94.15	97.46
Race	PG-FDD CVPR'24 [24]	Xception	11.13	4.78	94.52	97.66
Race	Fairadapter ICASSP'25 [10]	ViT-L/14	43.22	20.39	56.03	71.50
Race	RSEF-FDD arxiv'25 [18]	Xception	8.39	5.28	93.60	97.09
Race	Ding et al. (published)	Xception	9.29	4.35	94.86	97.71
Race	CES-FL A: ImageNet frozen [s17]	Xception	31.07	14.16	60.79	67.48
Race	CES-FL B: detection frozen + BCE [s17]	Xception	11.41	2.37	96.04	98.35
Race	CES-FL C: detection frozen + invariance [s17]	Xception	11.36	2.46	95.72	98.22
Race	CES-FL upgrade R0: frozen BCE [s17]	Xception	11.41	2.37	96.04	98.35
Race	CES-FL upgrade R1: partial BCE [s17]	Xception	11.51	2.95	95.98	98.40
Race	CES-FL upgrade R2: partial DRO [s17]	Xception	15.63	3.88	96.07	98.40
Race	CES-FL upgrade RN: no-change diagnostic [s17]	Xception	8.55	3.61	95.06	97.85
Race	CES-FL upgrade RD: age diagnostic [s17]	Xception	11.83	3.33	95.77	98.20
Intersection	Ori (published)	Xception	36.03	14.64	74.43	92.69
Intersection	DAG-FDD WACV'24 [20]	Xception	24.08	26.25	85.80	96.72
Intersection	DAW-FDD WACV'24 [20]	Xception	14.36	26.13	86.74	97.46
Intersection	PG-FDD CVPR'24 [24]	Xception	9.19	13.39	86.83	97.66
Intersection	Fairadapter ICASSP'25 [10]	ViT-L/14	86.91	42.44	45.98	71.50
Intersection	RSEF-FDD arxiv'25 [18]	Xception	23.64	21.74	85.77	97.09
Intersection	Ding et al. (published)	Xception	20.18	9.47	86.91	97.71
Intersection	CES-FL A: ImageNet frozen [s17]	Xception	61.82	25.20	53.13	67.48
Intersection	CES-FL B: detection frozen + BCE [s17]	Xception	37.89	11.84	90.90	98.35
Intersection	CES-FL C: detection frozen + invariance [s17]	Xception	37.48	12.10	90.30	98.22
Intersection	CES-FL upgrade R0: frozen BCE [s17]	Xception	37.89	11.84	90.90	98.35
Intersection	CES-FL upgrade R1: partial BCE [s17]	Xception	29.92	12.85	90.50	98.40
Intersection	CES-FL upgrade R2: partial DRO [s17]	Xception	30.23	12.26	90.57	98.40
Intersection	CES-FL upgrade RN: no-change diagnostic [s17]	Xception	22.93	13.03	90.55	97.85
Intersection	CES-FL upgrade RD: age diagnostic [s17]	Xception	26.27	12.94	90.09	98.20

Blue rows: our measured single-seed results. No best-method/significance claim. F_FPR is summed disparity, not max–min FPR gap. See README.md for group coverage, thresholds, hashes, caveats and latest VAL-only link.

Table 2. Cross-domain evaluation — Ding layout

Our rows: seed 17 TEST, stored validation-real FPR10% thresholds; no test recalibration. Published rows retain Ding's reported numbers. Shared format does not establish identical training/evaluation protocols.

A/B/C: earlier controlled pilot. R0/R1/R2/RN/RD: completed upgrade. C is invariance-only; RN/RD use six sources. Current 50+50 results are VAL only and excluded.

Datasets	Methods	Backbone	Gender			Race			Intersection			Overall
			F_FPR ↓	F_DP ↓	es-AUC ↑	F_FPR ↓	F_DP ↓	es-AUC ↑	F_FPR ↓	F_DP ↓	es-AUC ↑	AUC ↑
DFDC	Ori (published)	Xception	8.67	6.70	54.56	19.41	7.99	47.02	53.42	17.34	36.96	56.13
DFDC	DAG-FDD WACV'24 [20]	Xception	5.29	6.68	57.41	9.26	12.51	45.91	45.30	12.92	36.42	59.13
DFDC	DAW-FDD WACV'24 [20]	Xception	3.60	3.67	55.88	22.36	9.72	47.77	46.80	12.69	36.63	56.90
DFDC	PG-FDD CVPR'24 [24]	Xception	2.35	2.57	57.71	10.10	11.49	50.53	32.16	17.71	37.42	59.64
DFDC	Fairadapter ICASSP'25 [10]	ViT-L/14	2.39	4.57	55.88	32.49	26.77	52.27	74.10	31.53	31.39	59.25
DFDC	RSEF-FDD arxiv'25 [18]	Xception	2.09	3.83	57.90	12.24	7.74	50.27	38.89	12.18	37.72	59.08
DFDC	Ding et al. (published)	Xception	1.76	3.67	57.94	17.93	7.58	52.33	37.37	11.92	39.03	61.86
DFDC	CES-FL A: ImageNet frozen [s17]	Xception	7.11	1.79	46.98	9.79	9.44	39.33	35.94	15.94	32.06	50.63
DFDC	CES-FL B: detection frozen + BCE [s17]	Xception	3.37	3.73	58.05	15.73	5.12	49.82	41.43	14.26	40.29	59.47
DFDC	CES-FL C: detection frozen + invariance [s17]	Xception	1.99	1.42	58.84	13.18	7.12	51.33	56.31	22.84	42.87	59.67
DFDC	CES-FL upgrade R0: frozen BCE [s17]	Xception	3.37	3.73	58.05	15.73	5.12	49.82	41.43	14.26	40.29	59.47
DFDC	CES-FL upgrade R1: partial BCE [s17]	Xception	4.87	5.10	57.40	20.96	7.84	48.33	43.04	12.10	39.24	58.97
DFDC	CES-FL upgrade R2: partial DRO [s17]	Xception	6.68	7.88	58.63	22.18	8.04	50.24	41.53	14.52	42.16	59.52
DFDC	CES-FL upgrade RN: no-change diagnostic [s17]	Xception	5.24	7.23	59.51	19.50	7.62	49.29	40.69	13.72	40.22	59.67
DFDC	CES-FL upgrade RD: age diagnostic [s17]	Xception	7.83	8.77	58.49	20.28	6.75	50.26	37.84	14.63	42.86	59.86
Celeb-DF	Ori (published)	Xception	6.93	20.23	60.94	23.44	17.89	61.21	27.38	18.72	61.22	69.18
Celeb-DF	DAG-FDD WACV'24 [20]	Xception	8.70	14.25	66.23	19.44	13.26	62.61	23.60	13.80	63.08	72.29
Celeb-DF	DAW-FDD WACV'24 [20]	Xception	8.71	12.73	68.31	14.76	7.80	66.47	24.75	13.78	59.24	74.09
Celeb-DF	PG-FDD CVPR'24 [24]	Xception	7.95	16.29	69.47	23.99	12.47	67.78	23.71	13.30	62.63	74.24
Celeb-DF	Fairadapter ICASSP'25 [10]	ViT-L/14	8.59	26.84	56.84	46.14	36.35	58.34	47.03	35.69	54.77	64.94
Celeb-DF	RSEF-FDD arxiv'25 [18]	Xception	1.94	35.63	68.92	17.40	27.48	67.80	24.17	27.73	62.56	74.36
Celeb-DF	Ding et al. (published)	Xception	6.41	10.81	69.68	16.60	7.29	69.96	23.32	12.01	63.27	76.75
Celeb-DF	CES-FL A: ImageNet frozen [s17]	Xception	3.82	9.54	59.17	21.02	29.72	59.55	26.11	31.74	58.18	64.28
Celeb-DF	CES-FL B: detection frozen + BCE [s17]	Xception	7.71	24.92	63.85	31.36	24.42	50.57	33.23	25.31	50.24	74.33
Celeb-DF	CES-FL C: detection frozen + invariance [s17]	Xception	7.57	24.50	63.39	30.68	23.64	50.52	32.62	24.54	50.26	73.70
Celeb-DF	CES-FL upgrade R0: frozen BCE [s17]	Xception	7.71	24.92	63.85	31.36	24.42	50.57	33.23	25.31	50.24	74.33
Celeb-DF	CES-FL upgrade R1: partial BCE [s17]	Xception	7.77	24.22	63.87	32.79	24.09	51.48	36.48	27.15	51.19	74.21
Celeb-DF	CES-FL upgrade R2: partial DRO [s17]	Xception	9.18	23.61	64.48	31.14	23.00	51.26	34.41	25.23	51.11	74.67
Celeb-DF	CES-FL upgrade RN: no-change diagnostic [s17]	Xception	10.46	28.42	62.27	34.95	27.76	52.55	36.08	28.78	50.06	71.01
Celeb-DF	CES-FL upgrade RD: age diagnostic [s17]	Xception	8.19	25.71	63.46	36.92	26.59	52.65	43.68	32.14	50.23	72.80
DFD	Ori (published)	Xception	9.45	6.63	72.68	7.75	22.31	69.07	35.06	25.53	59.20	74.32
DFD	DAG-FDD WACV'24 [20]	Xception	5.41	5.60	75.87	2.56	21.34	70.14	33.00	23.02	61.09	76.29
DFD	DAW-FDD WACV'24 [20]	Xception	5.30	10.63	71.68	4.32	20.16	63.24	33.81	24.72	56.01	73.71
DFD	PG-FDD CVPR'24 [24]	Xception	5.05	1.86	76.37	5.79	19.31	74.81	28.00	23.93	66.32	80.70
DFD	Fairadapter ICASSP'25 [10]	ViT-L/14	6.32	7.55	63.28	11.29	12.28	56.28	29.56	33.11	48.63	68.12
DFD	RSEF-FDD arxiv'25 [18]	Xception	18.50	6.11	75.97	19.45	19.40	72.65	35.34	10.97	67.24	80.54
DFD	Ding et al. (published)	Xception	4.72	5.49	78.98	2.09	18.66	77.44	27.32	22.81	69.73	81.46
DFD	CES-FL A: ImageNet frozen [s17]	Xception	0.92	2.28	58.85	4.60	8.26	56.40	8.22	11.09	52.70	59.31
DFD	CES-FL B: detection frozen + BCE [s17]	Xception	6.20	1.96	81.97	8.61	0.05	81.36	39.63	5.32	73.86	82.36
DFD	CES-FL C: detection frozen + invariance [s17]	Xception	5.57	1.79	80.66	6.00	1.94	80.49	37.38	6.00	71.50	81.24
DFD	CES-FL upgrade R0: frozen BCE [s17]	Xception	6.20	1.96	81.97	8.61	0.05	81.36	39.63	5.32	73.86	82.36
DFD	CES-FL upgrade R1: partial BCE [s17]	Xception	5.88	1.78	81.23	10.21	0.58	80.55	39.62	4.80	72.48	81.81
DFD	CES-FL upgrade R2: partial DRO [s17]	Xception	5.96	2.04	81.91	9.91	0.66	81.14	43.29	4.76	73.28	82.12
DFD	CES-FL upgrade RN: no-change diagnostic [s17]	Xception	3.74	2.12	80.62	6.75	1.20	80.82	34.27	6.66	73.06	81.53
DFD	CES-FL upgrade RD: age diagnostic [s17]	Xception	5.59	1.43	82.00	10.89	1.06	81.26	36.58	4.13	76.52	82.28

Blue rows: our measured single-seed results. No best-method/significance claim. F_FPR is summed disparity, not max–min FPR gap. See README.md for group coverage, thresholds, hashes, caveats and latest VAL-only link.

Table 3. ResNet-50 backbone evaluation — Ding layout

Our rows: seed 17 TEST, stored validation-real FPR10% thresholds; no test recalibration. Published rows retain Ding's reported numbers. Shared format does not establish identical training/evaluation protocols.

A/B/C: earlier controlled pilot. R0/R1/R2/RN/RD: completed upgrade. C is invariance-only; RN/RD use six sources. Current 50+50 results are VAL only and excluded.

Datasets	Methods	Backbone	Gender			Race			Intersection			Overall
			F_FPR ↓	F_DP ↓	es-AUC ↑	F_FPR ↓	F_DP ↓	es-AUC ↑	F_FPR ↓	F_DP ↓	es-AUC ↑	AUC ↑
FF++	Ori (published)	ResNet-50	8.90	9.27	90.40	17.49	7.04	89.17	38.26	15.43	76.14	94.29
FF++	DAG-FDD WACV'24 [20]	ResNet-50	1.92	6.02	91.09	6.38	3.98	91.39	32.58	11.06	80.64	94.74
FF++	DAW-FDD WACV'24 [20]	ResNet-50	3.25	5.29	90.13	12.46	3.29	90.92	31.07	11.03	80.98	93.96
FF++	PG-FDD CVPR'24 [24]	ResNet-50	3.05	3.06	93.57	10.89	2.83	91.53	22.10	9.69	81.58	94.46
FF++	Ding et al. (published)	ResNet-50	3.94	5.14	94.46	18.67	2.43	92.69	34.30	8.61	82.87	95.53
FF++	CES-FL A: ImageNet frozen [s17]	ResNet-50	2.96	2.42	70.46	21.53	9.14	62.72	52.22	19.88	49.69	71.13
FF++	CES-FL B: detection frozen + BCE [s17]	ResNet-50	1.01	6.22	93.46	21.78	8.68	92.23	48.99	15.66	82.63	95.98
FF++	CES-FL C: detection frozen + invariance [s17]	ResNet-50	0.96	6.24	93.53	20.11	8.67	92.32	46.79	15.74	82.79	95.89
DFDC	Ori (published)	ResNet-50	12.55	11.06	54.96	17.34	8.38	48.30	48.10	16.30	35.76	57.83
DFDC	DAG-FDD WACV'24 [20]	ResNet-50	6.57	4.87	57.63	15.96	2.95	50.31	44.10	11.25	33.41	60.29
DFDC	DAW-FDD WACV'24 [20]	ResNet-50	9.12	7.14	57.22	14.60	2.81	48.94	43.31	9.55	34.71	59.78
DFDC	PG-FDD CVPR'24 [24]	ResNet-50	5.78	4.65	58.34	12.68	5.23	49.33	31.97	8.54	34.85	61.83
DFDC	Ding et al. (published)	ResNet-50	3.97	3.61	56.96	11.94	5.15	50.99	32.14	10.27	37.55	62.54
DFDC	CES-FL A: ImageNet frozen [s17]	ResNet-50	6.50	5.65	55.15	3.92	6.85	48.30	29.22	10.75	35.31	56.41
DFDC	CES-FL B: detection frozen + BCE [s17]	ResNet-50	5.21	7.18	57.71	27.35	11.17	49.51	53.30	17.06	34.98	58.07
DFDC	CES-FL C: detection frozen + invariance [s17]	ResNet-50	4.62	6.70	57.91	26.81	10.16	49.63	52.17	15.38	34.86	58.37
Celeb-DF	Ori (published)	ResNet-50	12.26	22.12	61.11	18.55	22.43	59.82	24.84	23.18	56.09	71.65
Celeb-DF	DAG-FDD WACV'24 [20]	ResNet-50	8.22	20.45	67.01	23.07	18.50	68.07	20.29	19.33	66.76	75.77
Celeb-DF	DAW-FDD WACV'24 [20]	ResNet-50	9.03	20.85	68.33	18.60	17.65	69.14	23.84	18.55	68.74	76.51
Celeb-DF	PG-FDD CVPR'24 [24]	ResNet-50	6.98	16.92	68.49	16.74	16.50	70.35	29.34	18.72	66.26	76.10
Celeb-DF	Ding et al. (published)	ResNet-50	4.53	12.00	68.98	6.76	8.16	71.03	5.56	9.11	68.25	77.35
Celeb-DF	CES-FL A: ImageNet frozen [s17]	ResNet-50	7.68	7.08	62.14	18.47	11.23	56.60	20.05	19.12	59.50	66.44
Celeb-DF	CES-FL B: detection frozen + BCE [s17]	ResNet-50	9.46	26.73	63.16	29.76	26.39	58.64	30.27	27.33	57.40	72.80
Celeb-DF	CES-FL C: detection frozen + invariance [s17]	ResNet-50	9.63	27.54	63.13	31.36	27.40	59.78	31.85	28.35	58.62	72.81
DFD	Ori (published)	ResNet-50	10.95	9.08	70.33	4.66	12.57	66.68	33.46	16.84	56.00	71.31
DFD	DAG-FDD WACV'24 [20]	ResNet-50	3.62	4.19	73.74	3.86	13.54	70.33	25.22	14.84	57.19	73.97
DFD	DAW-FDD WACV'24 [20]	ResNet-50	4.81	3.90	72.12	7.36	3.21	70.57	33.45	14.30	54.52	73.58
DFD	PG-FDD CVPR'24 [24]	ResNet-50	3.57	1.82	76.14	2.62	7.81	72.16	23.15	11.01	62.65	77.47
DFD	Ding et al. (published)	ResNet-50	3.12	1.76	77.68	6.51	3.03	70.93	20.99	4.85	66.19	78.63
DFD	CES-FL A: ImageNet frozen [s17]	ResNet-50	4.53	0.51	47.31	8.28	0.32	42.13	21.41	15.92	35.78	50.20
DFD	CES-FL B: detection frozen + BCE [s17]	ResNet-50	12.60	5.94	71.41	10.06	1.70	71.34	40.38	14.18	54.45	73.91
DFD	CES-FL C: detection frozen + invariance [s17]	ResNet-50	13.01	6.20	71.20	9.76	1.52	71.72	39.62	15.06	54.59	74.02

Blue rows: our measured single-seed results. No best-method/significance claim. F_FPR is summed disparity, not max–min FPR gap. See README.md for group coverage, thresholds, hashes, caveats and latest VAL-only link.